

Technical validation (working document)

Liza Buriak^{1*}

^{1*}Institute for Ecological Economics, Vienna University of Economics and Business (WU), Vienna, Austria.

Corresponding author(s). E-mail(s): TODO@wu.ac.at;

Abstract

Standalone draft of the Technical validation section, merging the Validation and uncertainty subsection of the Methods working document (methods.v2), the standalone Validation draft, and the Uncertainty and sensitivity appendix. The text is self-contained; pointers to the Methods appendices are given in plain text.

1 Technical validation

1.1 Agreement with the Trase benchmark

The validation targets the subnational half of the pipeline. How much soy each country imports from Brazil, and how it propagates through global supply chains to final consumers, is carried by FABIO, a peer-reviewed global physical input-output database whose construction and evaluation are documented elsewhere [1]; the pipeline adds no new information at that level. What is new, and therefore what has to be validated, is the subnational layer: which municipalities the exported soy comes from and how it is routed to each importer.

The reconstructed origin-to-importer flows are evaluated against the independent Trase Brazil-soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [2, 3]. Agreement is measured by the Pearson correlation between the modelled and the benchmark flows, computed over all matched municipality–destination pairs of a year (pooled), for the pairs of one importing country at a time (per-destination), or on $\log_{10}$ flows in the rocket plots of Section 1.2. Two features organise the comparison. First, agreement depends on flow size: the model and the benchmark agree closely on the large flows that carry the tonnage and diverge on the small ones, which are numerous but negligible in volume (Section 1.2). Second, from 2019 onward the comparison target itself changes: Trase switched its per-shipment source, and part of the post-2018 disagreement originates in the Trase data rather than in the model (Section 1.2).

The dependence on volume shapes the choice of the headline metric. Nearly half of the matched flows (46%) are below 1,000 t, and together they carry 1.3% of the volume (Figure 2); a plain average over destinations therefore lets the noise of negligible flows dominate, giving Cuba the same weight as China. The headline metric is instead the volume-weighted per-destination correlation: for each importing country the modelled map of origin municipalities is correlated with the map Trase reports for that importer, and the per-importer scores are averaged with the importer’s share of the year’s volume as weight (Figure 1).

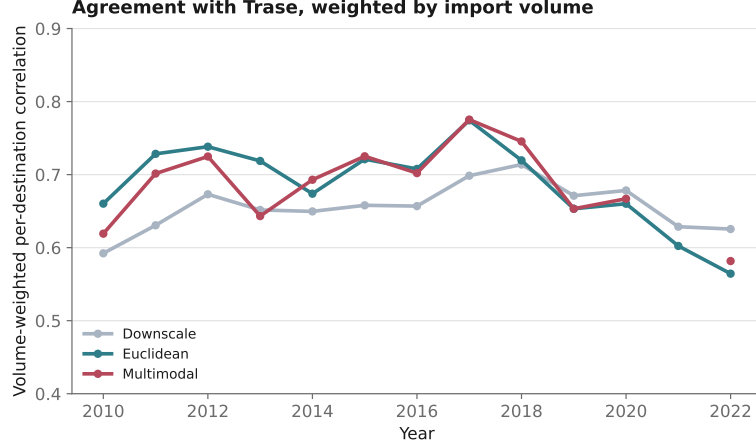


Fig. 1 Volume-weighted per-destination correlation with the Trase benchmark, 2010–2022, for the three allocation variants (the multimodal variant has no 2021 solve). Weights are each importer’s share of the year’s Trase-recorded import volume; Brazil (domestic use) is excluded. The supply-chain models exceed the downscaling baseline in every year up to 2018; after the benchmark’s 2019 change of source the ordering inverts (Section 1.2).

All three allocation variants are compared on this metric: the downscaling baseline, in which every importer draws from the same municipal map and only totals differ, the Euclidean model, and the multimodal transport variant. The weighted correlation of the Euclidean model runs between 0.66 and 0.77 over 2010–2018 and exceeds the baseline in every year of that period, by 0.05 to 0.10 in most years, with the gap closing in 2018; the multimodal variant tracks the Euclidean model within 0.03 in most years and 0.08 at worst (2013). Since import totals are identical in all variants, this gap is exactly the information added by the supply-chain structure: different importers source from different parts of Brazil, and only the supply-chain models reproduce such differences. After 2018 all variants decline and the ordering inverts, which Section 1.2 traces to the benchmark’s change of source.

The average conceals large differences between destinations. China, which takes most of the exported tonnage, scores 0.69 to 0.86 across years and Spain reaches 0.93, while for the smallest importers the scores scatter widely from year to year, from slightly negative to above 0.9, with no persistence. The validation appendix of the Methods document reports the full per-importer trajectories for the four importers that cover 70% of the volume and for the five smallest importers matched in every year.

Part of the distance that remains to the benchmark is a data limitation rather than a model choice. Installed crushing capacity is published by ABIOVE as state totals together with a roster of plant locations, with no capacities for individual plants, so the pipeline has to fix an allocation rule and divides each state’s capacity equally among its plants; Section 1.3 quantifies what this assumption costs. Trase, by contrast, observes individual shipments: per-shipment customs records name the exporting company, and the company’s tax registration identifies its silos and crushing facilities and their municipalities, so most of each flow is anchored to a specific logistics hub before any modelling; only the last step, from the hub to the municipality of production, is filled by a cost-minimising allocation similar in spirit to ours [4]. Per-shipment records of that granularity are proprietary and not available to an open pipeline. Sample construction is specified in the validation appendix of the Methods document.

1.2 Agreement grows with flow size

Agreement rises with flow size (Figure 2). Pooled over the 91,873 matched municipality×destination×year flows of 2010–2022, which span six orders of magnitude, the correlation of log flows with Trase is 0.68: large flows converge onto the 1:1 diagonal, small flows scatter. The same gradient shows in the per-destination scores:

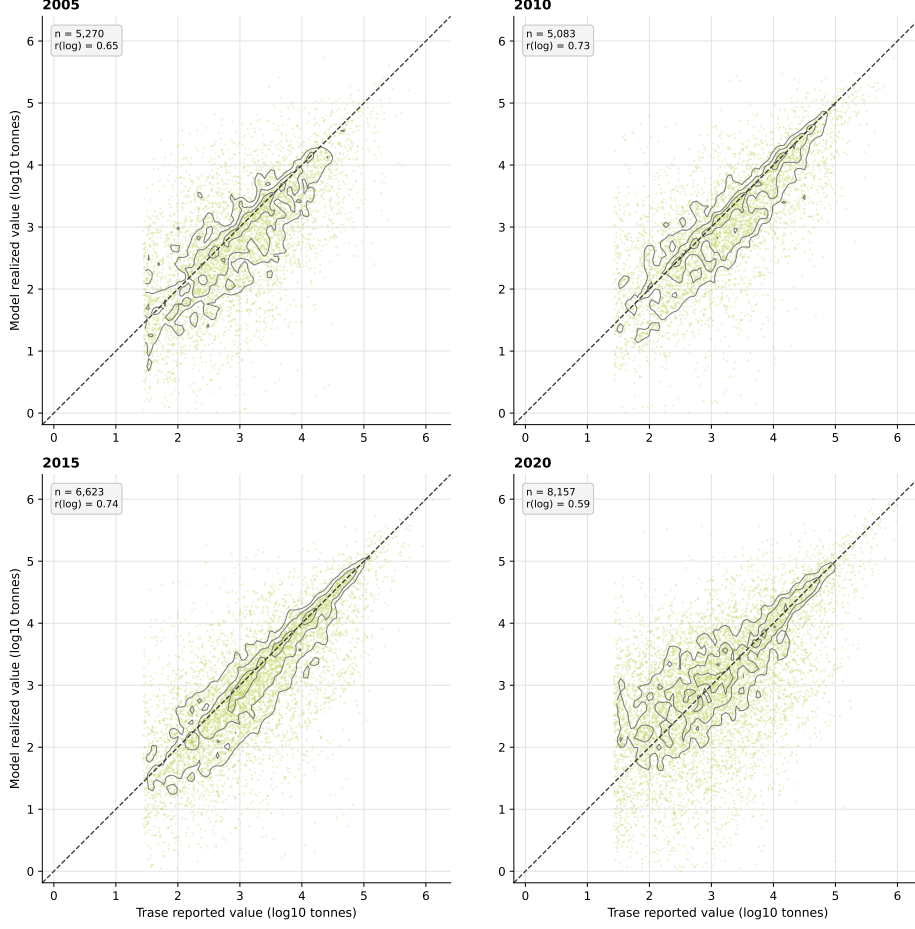


Fig. 2 Agreement with the benchmark grows with flow size (“rocket plots”). Every modelled municipality×destination soy export flow is plotted against the corresponding Trase flow (both $\log_{10}$ tonnes; flows below 1 t on either side are excluded) for four sample years. Because thousands of points overlap, grey contour lines outline where the point cloud is densest, in the way elevation lines outline a hill; the dashed line is the 1:1 diagonal on which a perfectly reproduced flow would lie. Each panel reports the number of matched flows and the log-scale Pearson correlation. Large flows converge onto the 1:1 diagonal (the “nose”), while small flows scatter (the “plume”).

the mean per-destination correlation rises from 0.29 in the lower half of import volumes to 0.53 in the upper half.

The weaker agreement visible in the 2020 panel is not specific to that year but marks the whole 2019–2022 block, and its origin is on the benchmark side. From 2019 onward Trase switched its per-shipment source to maritime-only bills of lading that record the country of first import rather than the final destination, and the share of its exports with unknown origin municipality roughly doubles [4]. At the switch the log-scale correlation steps down from 0.70–0.74 (2010–2017) to 0.59–0.65 (2019–2022), and the small flows drift above the diagonal: below 1,000 t the model reports on average 1.6 times the benchmark value in 2019–2020. The destination pattern identifies the mechanism. China, served by direct shipments, is unaffected (mean log deviation 0.00 in 2020), while Slovenia, whose port of Koper is a transshipment hub for landlocked Europe, is the strongest outlier in the opposite direction: the benchmark books to the country of first import the tonnage that the customs-based model books to final destinations, which in turn appear mildly over-reported (Thailand, Italy, Germany). Trase itself recommends against comparing its 2004–2018 and 2019–2022 periods directly, and the post-2018 part of the correlation band should be read with that caveat.

1.3 Sensitivity to input assumptions

The sensitivity analysis tests the two input assumptions that the data pin down least, and where the pipeline therefore has to assume the most: (a) the within-state location of crushing: ABIOVE reports capacity only as state totals, and the baseline divides each state’s total equally among its plants; (b) the feed and livestock parametrisation: the GLEAM per-system feed rates and the 2010-frozen municipal herd composition. Each perturbed variant re-runs the full flow reconstruction up to the origin-to-importer flows for a representative year (2019) and is scored against Trase on the pooled municipality $\times$ destination correlation (Euclidean model, base $r = 0.72$); twenty-four variants are run in total. The exact perturbation recipes are given in the validation appendix of the Methods document.

For assumption (a), the capacity map is perturbed in three ways:

1. production-proportional split: each state’s capacity is redistributed across its plant municipalities in proportion to municipal soybean production (1 run);
2. random splits: individual plant capacities are unobserved, so every plant is given a positive random weight, drawn from an exponential distribution, and each state’s fixed total is divided among its plants in proportion to those weights; this makes every possible split of a state’s total equally likely, and repeating it with ten different random seeds gives ten alternative capacity maps (10 runs);
3. allocation sharpness: municipal capacities act as weights that divide the national crush total among municipalities. Under the equal split they are identical within a state but still differ between states and where one municipality hosts several plants; raising every capacity to a power γ varies how strongly processing follows those remaining differences, with $\gamma = 2$ concentrating it in the highest-capacity municipalities and $\gamma = 0.5$ spreading it more evenly (2 runs).

For assumption (b), the parametrisation is perturbed in two ways:

1. random feed rates: the per-head bean and cake feed rates of every livestock system are multiplied by random factors, drawn independently for each system from a lognormal distribution with $\sigma = 0.3$, so that the factors typically lie between 0.75 and 1.35; repeating this with ten different random seeds gives ten parametrisations in which the GLEAM rates are wrong by realistic amounts in different directions (10 runs). The factors average to one and national feed totals are rescaled to the FAO commodity balance afterwards, so an error applied uniformly to all systems cancels exactly; what this perturbation tests is the system mix, since an error there shifts feed demand toward the municipalities whose herds are rich in the systems whose feed rates were scaled up in that draw;
2. herd composition: the 2010-frozen municipal herd composition is replaced by state-average system shares (1 run).

The two assumptions separate cleanly (Table 1). Crush location accounts for nearly all of the induced variation: the random within-state splits span correlations of 0.67 to 0.74 (spread 0.073), the production-proportional split scores 0.69, and the allocation-sharpness variants span 0.024. The feed variants are immaterial: the ten feed-rate draws move the correlation by less than 0.01 and the herd-composition variant by less than 0.001.

1.4 Backend consistency over time

Before 2010 the pipeline runs on FABIO v1.1 inputs instead of the v2 build. In the clean overlap years the two backends differ by 6–8% in the total footprint with cross-country correlations above 0.988, so pre-2010 levels are reported unadjusted.

1.5 Summary

Taken together, the checks bound the model on every side on which it can be tested. The reconstructed flows agree with the independent Trase benchmark at a

Table 1 Input-assumption sensitivity of the model–Trase correlation (pooled municipality×destination flows, Euclidean model, 2019; base $r = 0.723$).

Assumption	Perturbation	Runs	r range	Spread
Crush location	production-proportional split	1	0.691	–
Crush location	random within-state splits	10	0.671–0.744	0.073
Crush location	allocation sharpness $\gamma \in \{0.5, 2\}$	2	0.706–0.731	0.024
Feed rates	random per-system factors (lognormal, $\sigma = 0.3$)	10	0.717–0.725	0.007
Herd composition	state-average system shares	1	0.722	–

volume-weighted per-destination correlation of 0.66 to 0.77 over 2010–2018, and the supply-chain structure adds information exactly where it should, at the destination level, where it exceeds a structure-free baseline in every year of that period. Agreement is strongest where the tonnage is: the flows that carry the footprint are the best-validated part of the distribution, with China scoring 0.69 to 0.86 across years. The reconstruction is stable under perturbation of its weakest inputs: across twenty-four perturbed re-runs the benchmark correlation stays between 0.67 and 0.74 around a base of 0.72, the feed and herd assumptions are immaterial, and the residual sensitivity is concentrated in a single known data gap, the within-state location of crushing capacity, which per-plant capacity data would close. The remaining modelling choices are equally bounded: the multimodal transport variant performs on par with the straight-line model, and the FABIO backend vintage shifts pre-2010 levels by 6–8% while leaving the cross-country pattern intact. Results for large flows and large importers can therefore be used with confidence, while results for individual small-volume destinations should be read with the caution that Section 1.2 quantifies.

References

- [1] Bruckner, M. *et al.* FABIO—the construction of the food and agriculture biomass input–output model. *Environmental Science & Technology* **53**, 11302–11312 (2019).
- [2] Godar, J., Persson, U. M., Tizado, E. J. & Meyfroidt, P. Towards more accurate and policy relevant footprint analyses: Tracing fine-scale socio-environmental impacts of production to consumption. *Ecological Economics* **112**, 25–35 (2015).
- [3] Escobar, N. *et al.* Spatially-explicit footprints of agricultural commodities: Mapping carbon emissions embodied in Brazil’s soy exports. *Global Environmental Change* **62**, 102067 (2020).
- [4] Trase. SEI-PCS Brazil soy v2.6 supply chain map: Data sources and methods. Trase, January 2025 revision (2025).